

Radio Astronomy - Line Emission

- two-level (radiative + collisional)

$$\epsilon_\nu = \frac{h\nu_0}{4\pi} \frac{n_l(C_{lu} + M_{lu})}{(A_{ul} + C_{ul} + M_{ul})} \phi(\nu)$$

$$\epsilon_\nu = \frac{h\nu_0}{4\pi} \frac{g_u n_l}{g_l} \frac{\left(\frac{I_\nu}{2h\nu_0^3/c^2} + \frac{C_{ul}}{A_{ul}} e^{-h\nu_0/kT_K} \right)}{\left(1 + \frac{I_\nu}{2h\nu_0^3/c^2} + \frac{C_{ul}}{A_{ul}} \right)} \phi(\nu)$$

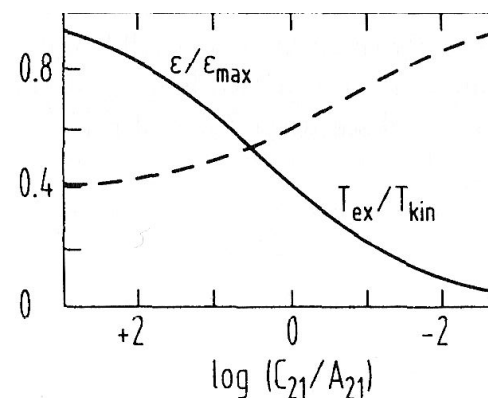
$$\frac{n_u}{n_l} = \frac{g_u}{g_l} e^{-h\nu_0/kT_{ex}}$$

$$\frac{C_{lu}}{C_{ul}} = \frac{g_u}{g_l} e^{-h\nu_0/kT_K}$$

$$\frac{I_\nu}{I_\nu + \frac{2h\nu_0^3}{c^2}} = e^{-h\nu_0/kT_b}$$

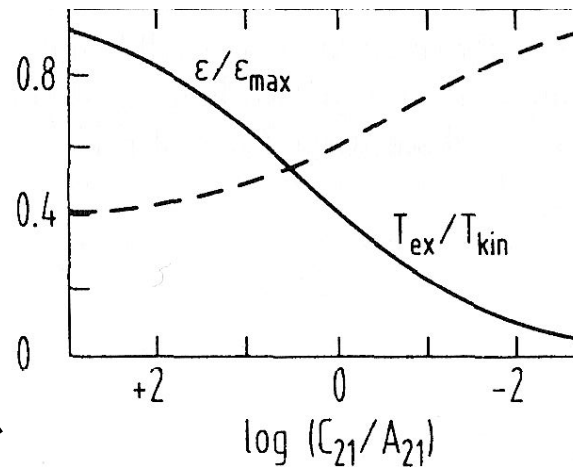
$$T_{ex} = T_K \frac{T_b A_{ul} + T_0 C_{ul}}{T_K A_{ul} + T_0 C_{ul}}$$

$$(T_{ex}, T_K, T_b \gg h\nu_0/k)$$



Radio Astronomy - Line Emission

- two-level (radiative + collisional)



LTE

$$I = B(T_b) = B(T)$$

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} e^{-h\nu_0/kT_{\text{ex}}} = \frac{g_u}{g_l} e^{-h\nu_0/kT}$$

$$T_b = T_{\text{ex}} = T_K = T$$

**in general
Non-LTE**

$$T_b < T_{\text{ex}} < T_K$$

(but,) no masing

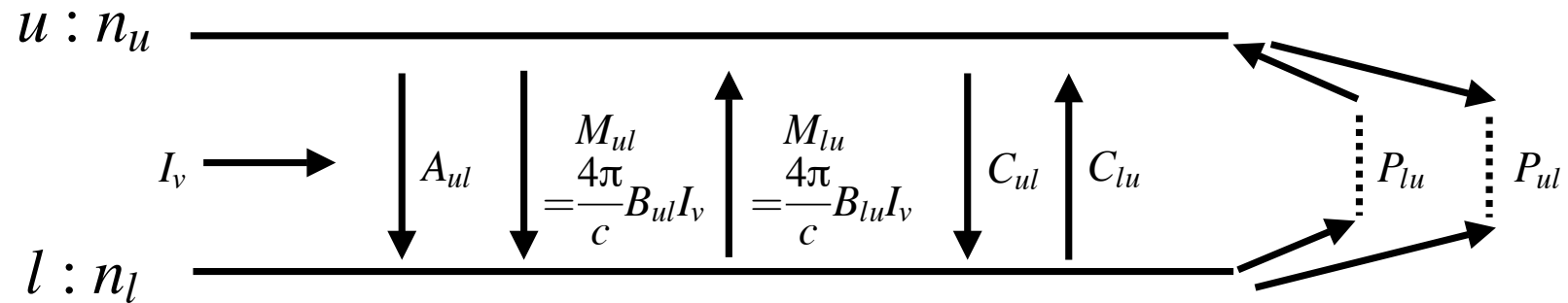
What is critical density?

Radio Astronomy - Line Emission

- three-level (radiative + collisional) or “two-level” + “pumping”

$$\frac{dI_\nu}{ds} = -\kappa_\nu I_\nu + \varepsilon_\nu$$

$$\frac{dI_\nu}{ds} = \frac{h\nu_0}{4\pi} \left[(n_u - n_l) B \frac{4\pi}{c} I_\nu + n_u A \right] \phi(\nu)$$



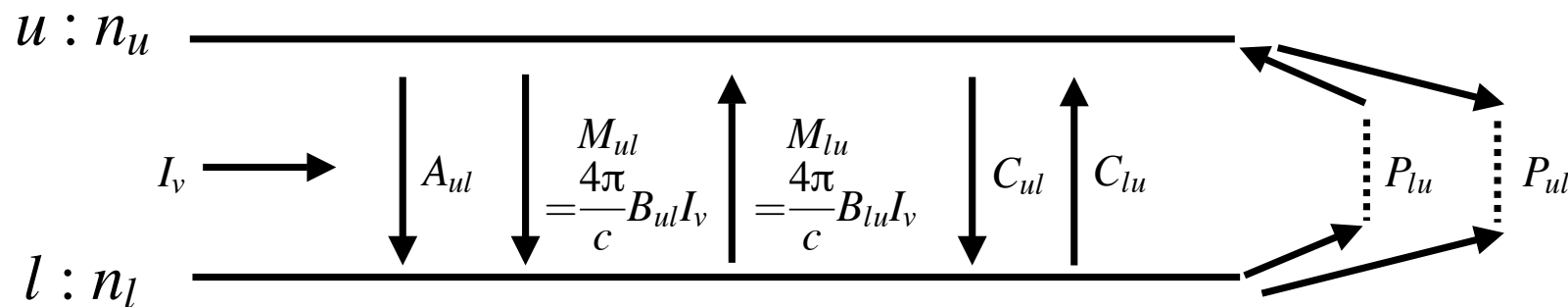
LTE $\frac{n_u}{n_l} = \frac{g_u}{g_l} e^{-h\nu_0/kT_K}$

Radio Astronomy - Line Emission

- three-level (radiative + collisional) or “two-level” + “pumping”

$$\frac{dI_\nu}{ds} = -\kappa_\nu I_\nu + \varepsilon_\nu$$

$$\frac{dI_\nu}{ds} = \frac{h\nu_0}{4\pi} \left[(n_u - n_l) B \frac{4\pi}{c} I_\nu + n_u A \right] \phi(\nu)$$



Non-LTE $n_u(A_{ul} + C_{ul} + M_{ul} + P_{ul}) = n_l(C_{lu} + M_{lu} + P_{lu})$

Radio Astronomy - Line Emission

- Steady State [Statistical Equilibrium]

$$n_u(A_{ul} + C_{ul} + M_{ul} + P_{ul}) = n_l(C_{lu} + M_{lu} + P_{lu})$$

For simplification, assume

$$\begin{aligned} C_{ul} &\approx C_{lu} \approx C \\ M_{ul} &\approx M_{lu} \approx M \\ A &\ll C, A \ll M \end{aligned}$$

$$\frac{n_u}{n_l} = \frac{P_{lu} + M + C}{P_{ul} + M + C}$$

For more simplification, assume $g_u = g_l$

in a special case

$$\begin{aligned} (n_u - n_l)|_{M=C=0} &= (n_u + n_l) \frac{P_{lu} - P_{ul}}{P_{lu} + P_{ul}} \\ &= n \frac{P_{lu} - P_{ul}}{P_{lu} + P_{ul}} \\ &\equiv \Delta n_0 \end{aligned}$$

in a more general case

$$\Delta n = \frac{\Delta n_0}{1 + \frac{2(C+M)}{P}}$$

Radio Astronomy - Line Emission

- Population Inversion

$$\Delta n = \frac{\Delta n_0}{1 + \frac{2(C+M)}{P}}$$

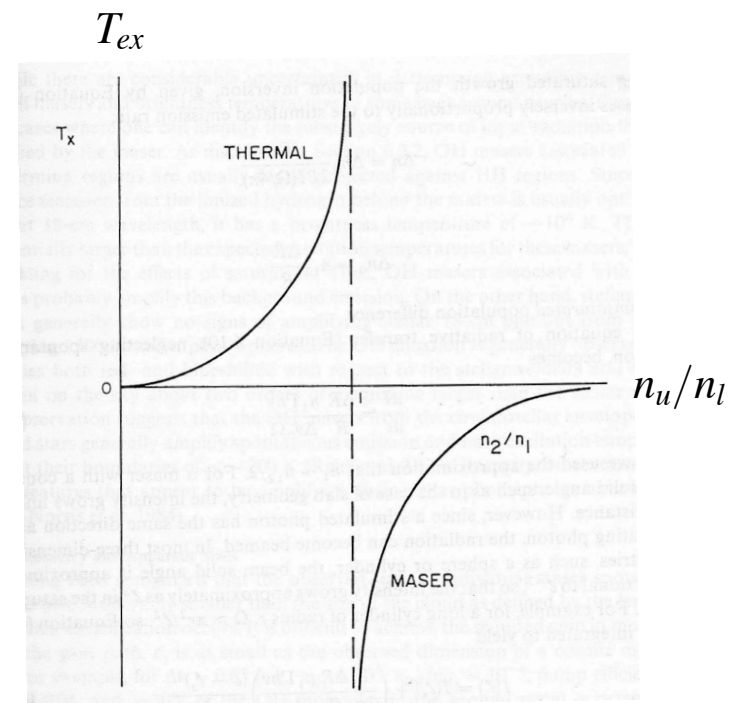
$$\text{If } \Delta n_0 > 0 \Rightarrow \Delta n > 0$$

$$\frac{n_u}{n_l} = e^{-h\nu_0/kT_{ex}} > 1$$

$$\Downarrow$$

$$T_{ex} < 0$$

excitation temperature



Radio Astronomy - Line Emission

- Amplification [in transfer equation]

$$\frac{dI_\nu}{ds} = \frac{\alpha I_\nu}{1 + I_\nu/I_s} + \varepsilon$$

$$\alpha = \frac{h\nu_0}{c} B \frac{\Delta n_0}{1 + \frac{2C}{P}} \phi(\nu)$$

$$I_s = \frac{cP}{2B\Omega_m} \left(1 + \frac{2C}{P}\right)$$

$$\varepsilon = \frac{h\nu_0}{4\pi} n_u A \phi(\nu)$$

Unsaturated
exponential amplification

$$I_{\nu_0} = I_0 e^{\alpha L} + \frac{\varepsilon}{\alpha_0} (e^{\alpha_0 L} - 1)$$

$$T_b = T_c e^{\alpha L} + |T_{ex}| (e^{\alpha_0 L} - 1)$$

Saturated
linear amplification

$$I_{\nu_0} = I_0 + (\alpha_0 I_s + \varepsilon) L$$

Radio Astronomy - Line Emission

- Line width [in transfer equation] : unsaturated case

$$I_{\nu_0} = I_0 e^{\alpha L} + \frac{\epsilon}{\alpha_0} (e^{\alpha_0 L} - 1)$$

$$T_b = T_c e^{\alpha L} + |T_{ex}| (e^{\alpha_0 L} - 1)$$

consider background amplification

$$T_b(\nu) = T_c e^{\alpha(\nu)L}$$

$$\alpha(\nu) = \alpha_0 \exp\left(-\frac{(\nu - \nu_0)^2}{2\sigma_0^2}\right)$$

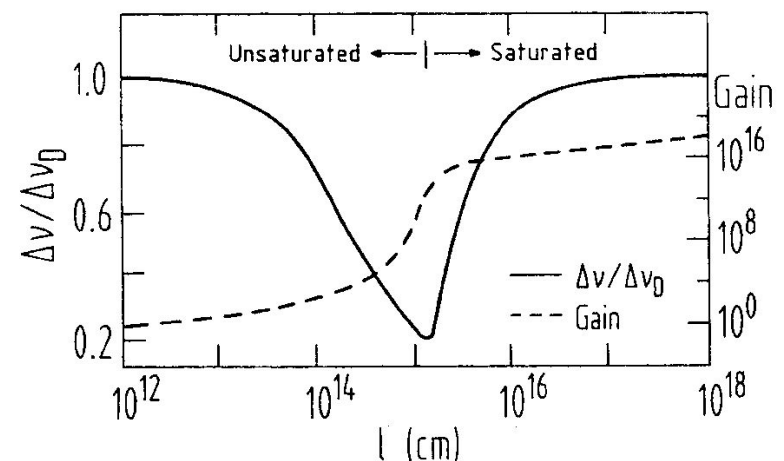
$$T_b(\nu) = T_c e^{\alpha_0 \exp\left(-\frac{(\nu - \nu_0)^2}{2\sigma_0^2}\right)L}$$

$$\alpha(\nu) \approx \alpha_0 \left(1 - \frac{(\nu - \nu_0)^2}{2\sigma_0^2}\right)$$

$$\begin{aligned} T_b(\nu) &= [T_c e^{\alpha_0 L}] e^{-\frac{\alpha_0 L}{2\sigma_0^2} (\nu - \nu_0)^2} \\ &= T_{\nu_0} e^{-\frac{\alpha_0 L}{2\sigma_0^2} (\nu - \nu_0)^2} \end{aligned}$$

line-width and path length

$$\sigma = \frac{\sigma_0}{\alpha_0 L}$$



Radio Astronomy - Masers

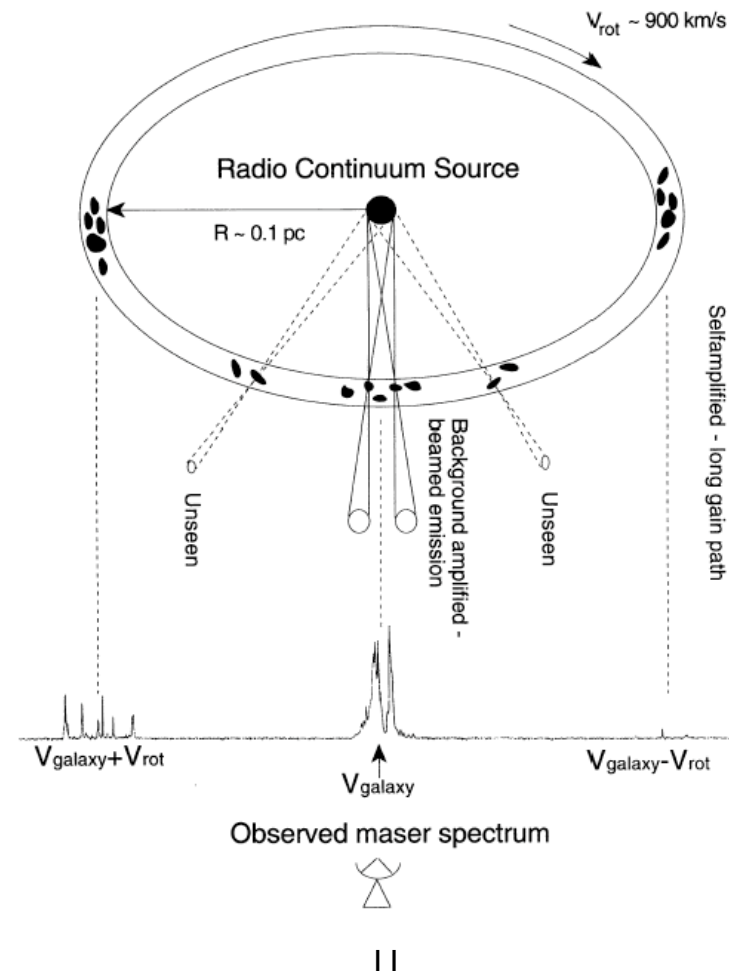
- Masers
 - physical condition required
 - population inversion : still mostly poor understood
 - collisional pumping
 - radiative pumping
 - physical information provided
 - density/temperature (required for pumping)
 - accurate velocity (required for pumping)
 - kinematic information on “small” parcel of gas
 - accurate (relative) positional information
 - proper motion measurements - distance

Radio Astronomy - Masers

- Masers
 - species
 - OH, H₂O, SiO, H₂CO, CH₃OH, HCN, NH₃, H-recomb
 - locations
 - Galactic
 - ISM : star forming regions, HII regions
 - OH, H₂O, SiO, H₂CO, CH₃OH, NH₃, H-recomb
 - stellar : circumstellar envelopes around evolved stars
 - SiO, OH, H₂O, HCN
 - solar : comets
 - Extragalactic
 - “megamaser” ($\sim 10^6$ times brighter than galactic counterparts), or even “gigamaser” for OH
 - H₂O in NGC 4945 (Santos and Lepine 1979)
 - OH in Arp 220 (Baan et al. 1982)
 - circumnuclear disk
 - amplification of radio continuum
 - OH (FIR?), H₂CO (FIR/Radio?), H₂O (collisional?)
 - star formation?

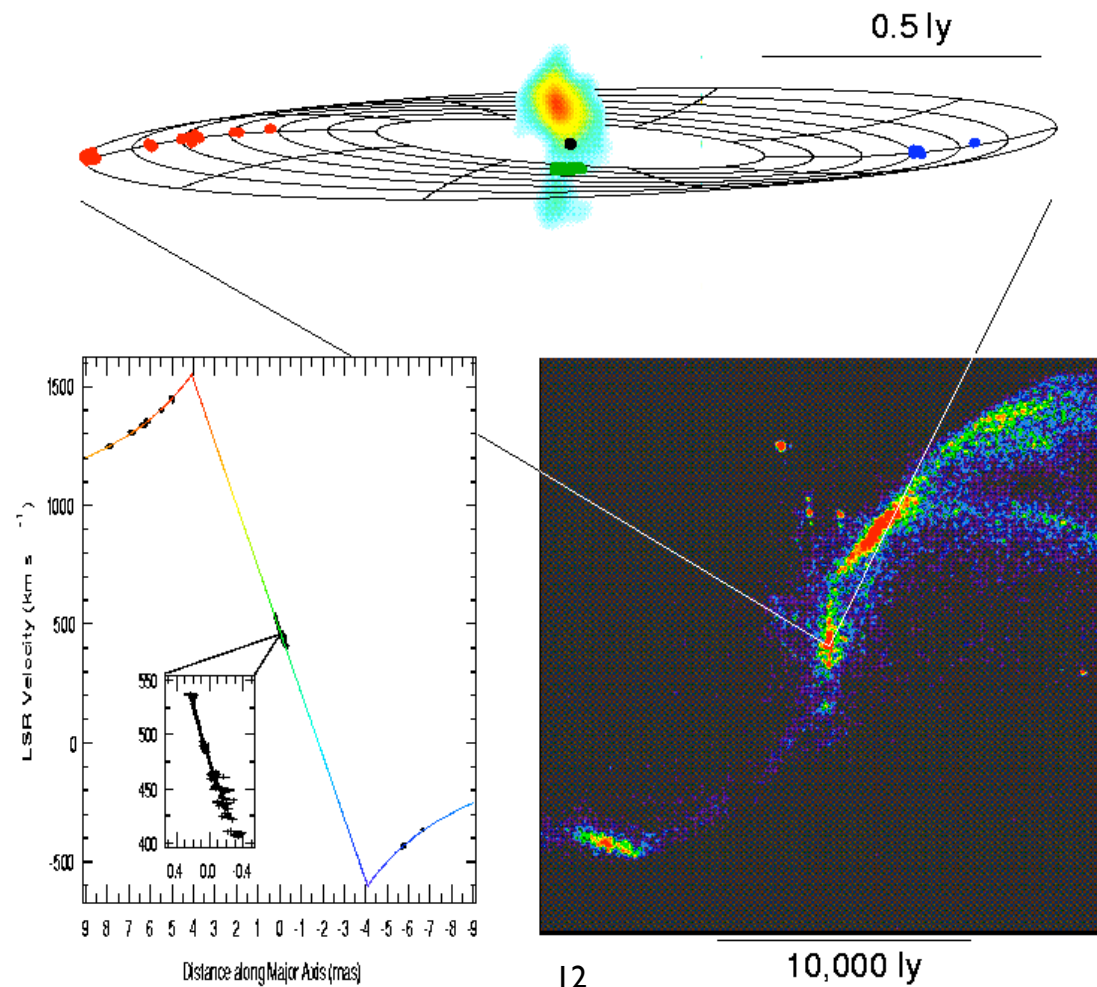
Radio Astronomy - Masers

- Megamasers : SMBH in NGC 4258 (Syfert-2 AGN)



Radio Astronomy - Masers

- Megamasers : SMBH in NGC 4258 (Syfert-2 AGN)



Radio Astronomy - The 21 cm Line

- The 21 cm line
 - first radio wavelengths spectral line detected
 - prediction : van de Hulst and Oort
 - detection : Ewen and Purcell
 - Hyperfine splitting - interaction between the magnetic moment of the proton and the magnetic moment of the orbiting electron at the ground state $1^2S_{1/2}$
- Einstein A ($2.86888 \times 10^{-15}/s$) due to
 - small (magnetic) dipole transition
 - low frequency
- Rest Frequency : 1.420405751786 +/- 0.000000000030 MHz

Radio Astronomy - The 21 cm Line

- The 21 cm line
 - excitation, general called “spin”, temperature

$$\frac{N_u}{N_l} = \frac{g_u}{g_l} \exp\left(\frac{-h\nu_0}{kT_e}\right)$$

$$= 3 \exp\left(\frac{-h\nu_0}{kT_s}\right)$$

- recall

$$T_s = T_K \frac{T_B A_{ul} + T_0 C_{ul}}{T_K A_{ul} + T_0 C_{ul}}$$

$$T_0 = \frac{h\nu_0}{k} = 0.0682K$$

$$= \frac{T_B + y T_K}{1 + y}$$

$$y = \frac{h\nu C_{ul}}{k T_K A_{ul}}$$

Purcell & Field (1956)

$$= \frac{T_K + y' T_B}{1 + y'}$$

$$y' = \frac{k T_K A_{ul}}{h\nu C_{ul}}$$

Kulkarni & Heiles (1988)

Radio Astronomy - The 21 cm Line

- The 21 cm line
 - absorption coefficient, optical depth, and column density

$$\kappa_\nu = \frac{3c^2}{32\pi\nu_{ul}} A_{ul} n_H \frac{h}{kT_s} \phi(\nu)$$

$$d\tau(\nu) = -\kappa_\nu(s, \nu) ds$$

$$d\tau\left(\frac{\nu}{\text{km s}^{-1}}\right) = -5.4873 \cdot 10^{-19} \left(\frac{n_H(s)}{\text{cm}^{-3}}\right) \left(\frac{T_s(s)}{\text{K}}\right)^{-1} \left(\frac{\phi(\nu)}{\text{km}^{-1}\text{s}}\right) d\left(\frac{s}{\text{cm}}\right)$$

$$\tau(\nu) = \int -\kappa_\nu(s, \nu) ds$$

$$\begin{aligned} \tau\left(\frac{\nu}{\text{km s}^{-1}}\right) &= -5.4873 \cdot 10^{-19} \left(\frac{T_s}{\text{K}}\right)^{-1} \left(\frac{\phi(\nu)}{\text{km}^{-1}\text{s}}\right) \int \left(\frac{n_H(s)}{\text{cm}^{-3}}\right) d\left(\frac{s}{\text{cm}}\right) \\ &= -5.4873 \cdot 10^{-19} \left(\frac{T_s}{\text{K}}\right)^{-1} \left(\frac{\phi(\nu)}{\text{km}^{-1}\text{s}}\right) \frac{N_H}{\text{cm}^{-2}} \end{aligned}$$

$$\begin{aligned} \int \tau(\nu) d\nu &= -5.4873 \cdot 10^{-19} \left(\frac{T_s}{\text{K}}\right)^{-1} \frac{N_H}{\text{cm}^{-2}} \int \left(\frac{\phi(\nu)}{\text{km}^{-1}\text{s}}\right) d\nu \\ &= -5.4873 \cdot 10^{-19} \left(\frac{T_s}{\text{K}}\right)^{-1} \frac{N_H}{\text{cm}^{-2}} \end{aligned}$$

$$\Downarrow$$

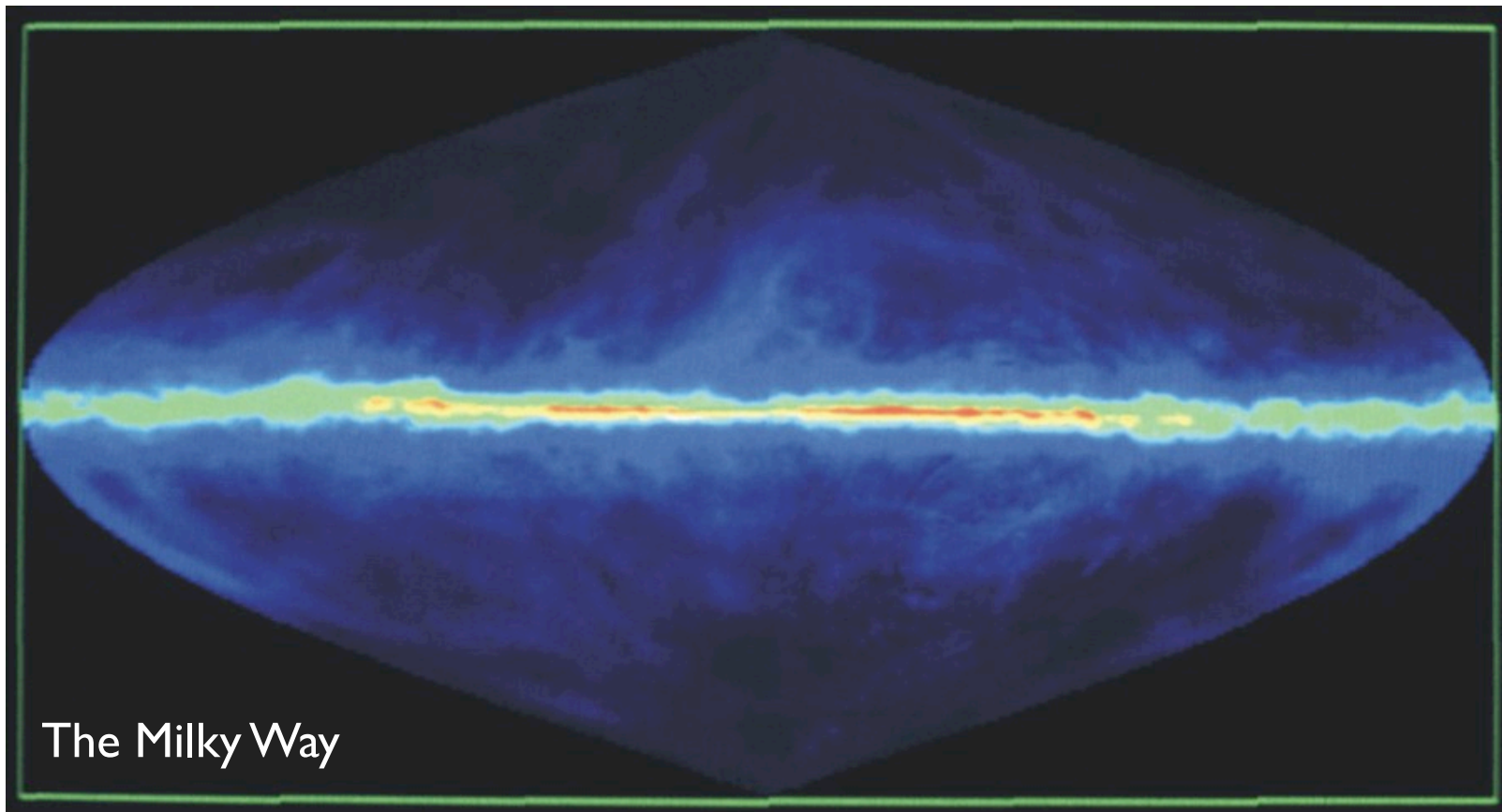
$$\frac{N_H}{\text{cm}^{-2}} = 1.8224 \cdot 10^{18} \left(\frac{T_s}{\text{K}}\right) \int \tau(\nu) d\frac{\nu}{\text{km s}^{-1}}$$



$$\begin{aligned} \tau &= -\ln\left(1 - \frac{T_L}{T_s - T_{BG}}\right) \\ T_s &\gg T_{BG} \\ &\approx -\ln\left(1 - \frac{T_L}{T_s}\right) \\ T_L &\ll T_s \text{ or } \tau \ll 1 \\ &\approx \frac{T_L}{T_s} \end{aligned}$$

Radio Astronomy - The 21 cm Line

- HI Clouds and the Galactic Structure



Radio Astronomy - The 21 cm Line

- HI Clouds and the Galactic Structure

- Interstellar Medium (ISM)

- gas heating and cooling

$$\frac{d}{dt} \left(\frac{3}{2} N k T_k \right) = \Gamma - \Lambda$$

- heating (ionization by radiation, high-energy particles [cosmic rays], collisions of gas clouds or streams)

- cooling (heavy element spectral emission)

- four component model (in pressure balance?) - the swiss cheese picture (with undetermined filling factors)

- CNM (cold neutral medium)

- narrow 21 cm absorption

- $T < 50\text{K}$, $n > 1 - 10 \text{ cm}^{-3}$

- WNM (warm neutral medium)

- ubiquitous wide 21 cm emission

- $T > 200\text{K}$

- WIM (warm ionized medium)

- widespread H α emission

- $T \sim 10^4\text{K}$

- HIM (hot ionized medium)

- diffuse soft X-ray emission and O VI absorption

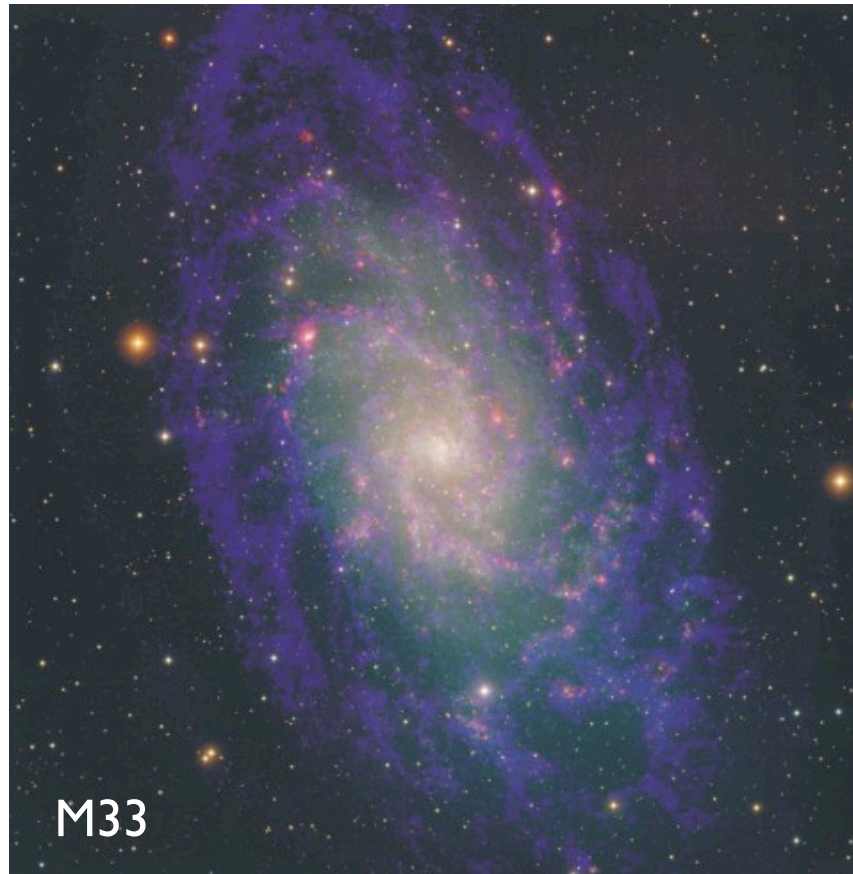
- $T \sim 10^6\text{K}$

Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies
 - extragalaxies
 - major tool for kinematic studies, such as rotation curve since :
 - tracing relative more extended regions since :
 - HI - neutral atomic gas
 - low excitation/density requirement
 - spectral (line) feature - kinematic (velocity) probe
 - findings :
 - HI distribution not centered
 - flat rotation curve
 - tidal interaction features

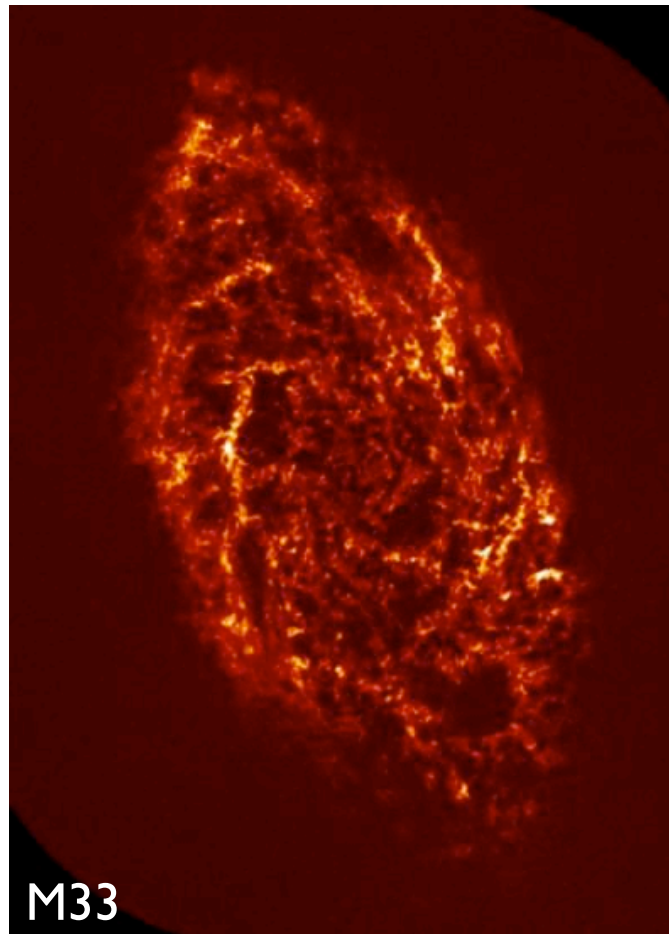
Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies



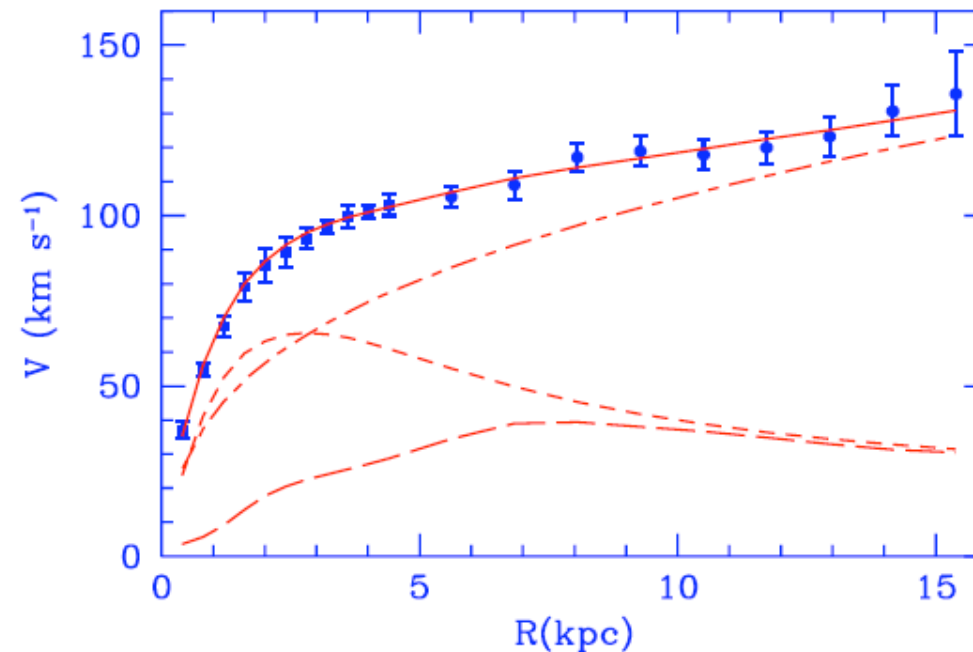
Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies



Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies



M33 rotation curve (points) compared with the best fit model (continuous line). Also shown the halo contribution (dashed-dotted line), the stellar disk (short dashed line) and the gas contribution (long dashed line) [Corbelli and Salucci 2000, MNRAS, 311, 441]

Radio Astronomy - The 21 cm Line

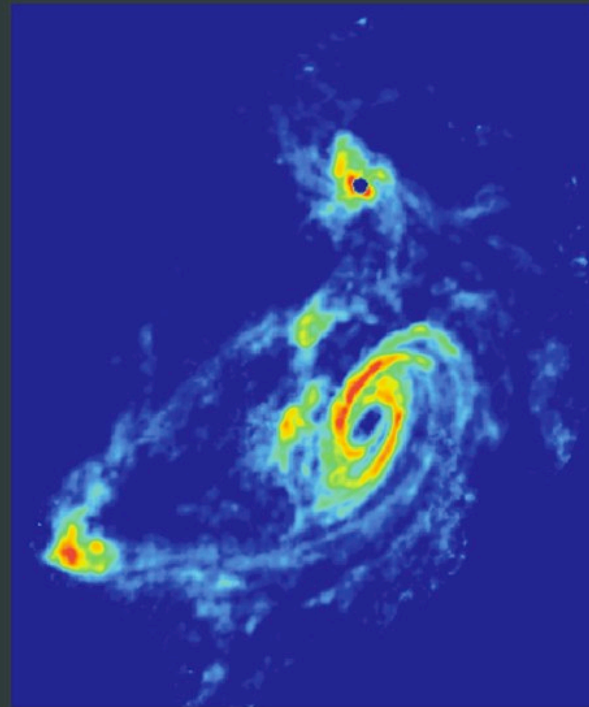
- HI in the extragalactic studies

TIDAL INTERACTIONS IN M81 GROUP

Stellar Light Distribution



21 cm HI Distribution



Radio Astronomy - The 21 cm Line

- HI in the extragalactic studies

